

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Mechanical Engineering) SEMESTER - 8 Summer 2025 (Regular)

Course :Bachelor of Technology (Mechanical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 8

Subject Code & Name: BTMEC802F - NON-CONVENTIONAL ENERGY RESOURCES

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

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- 1 Which of the following are energy conversion devices: 1. Turbine, 2. Nuclear Reactor, 3. Boiler 4. All components
- 2 Solar powered devices can work in night hours also. State true or false: 1) True, 2) False
- 3 State about battery and fuel cell _____. 1. Both are electrochemical devices, 2. Both can be used for electric vehicles, 3. Both options 1 and 2, 4. None of the options
- 4 Biomass implies burning of wood, animal or poultry wastes or organic matter. State true or false. 1) True, 2) False
- 5 Mechanical Energy is converted into Electrical Energy by the device. 1) Turbine, 2) Compressor, 3) Generator, 4) None of the components
- 6 The instrument used for measurement of global component of radiation of solar energy is. 1) Pyranometer, 2) Pyrheliometer, 3) Manometer, 4) All the instruments
- 7 What are the challenges with Ocean Thermal Energy Conversion plant? 1) Pressure, 2) Depth, 3) Water current, 4) All the options
- 8 Windmill's wind power depends on the parameters? 1) Volume of air going through it, 2) Cube power of wind speed, 3) Height of wind mill, 4) All the parameters
- 9 The pulverized coal is in the form of _____. : 1) Solid billets, 2) Powder, 3) Slurry, 4) Stones
- 10 Increase in GDP appears linked to increase in energy usage: State true or false. 1) True, 2) False
- 11 The working temperature range of solar concentrating collector is: 1) below 100 degree Celsius, 2) Above 300 degree Celsius
- 12 Gas turbine power plant comprises: 1) Compressor, 2) Turbine, 3) Combustion chamber, 4) All the options

Q2. Solve the following.

- A) Explain the working of Biogas power plant, with the help of neat sketches.

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- B) Describe the energy storage technology in the form of (i) flywheel and (ii) super-capacitor, with the help of neat line schematics. 6
- Q3. Solve the following.
- A) Discuss working principle and performance characteristics of solar photovoltaic power plant. 6
- B) Explain with the help of neat sketches the classification of wind turbines. 6
- Q4. Solve Any Two of the following.
- A) Explain the construction, working and testing procedures of liquid flat plate collector. 6
- B) Explain the working of Ocean Thermal Energy conversion plant with the help of neat schematic of components and cycle of operation. 6
- C) Discuss one of the instruments used for measurement of solar radiation, with the help of neat sketches. 6
- Q5. Solve Any Two of the following.
- A) State the classification of batteries & discuss with neat sketches the Li-Ion Battery. 6
- B) Describe the parameters which affect the performance of batteries and Photovoltaic panels. 6
- C) Describe the various types of Concentrating Collectors with the help of neat sketches. 6
- Q6. Solve Any Two of the following.
- A) Explain in brief about the attenuation of solar radiation received at the Earth's surface, mentioning spectral distribution of extra-terrestrial solar radiation and schematic representation of the mechanisms of absorption and scattering and beam and diffuse radiation received at the Earth's surface. 6
- B) Discuss in brief the working of 'Nuclear fuel power plant', with the help of neat line diagrams. 6
- C) What are fuel Cells? Explain operation principle of an acidic fuel cell. Mention the advantages of a fuel cell based power plant. 6

*** End ***